5	**(a)** **(i)**	vibrations (in plane) normal to direction of energy propagation	B1	[1]
	(ii)	vibrations in one direction (normal to direction of propagation)	B1	[1]
	(b) **(i)**	at (displacement) antinodes / where there are no heaps, wave has maximum amplitude (of vibration)	B1	
		at (displacement) nodes/where there are heaps, amplitude of vibration is zero/minimum	B1	
		dust is pushed to / settles at (displacement) nodes	B1	[3]
	(ii)	$2.5\lambda = 39$ cm	C1	
		$v = f\lambda$	C1	
		$v = 2.14 \times 10^{3} \times 15.6 \times 10^{-2}$		
		$= 334$ m s^{-1} (*allow 330, not 340*)	A1	[3]
	(c)	Stationary wave formed by interference / superposition / overlap of	B1	
		either wave travelling down tube and its reflection		
		or two waves of same (type and) frequency travelling in opposite directions	B1	
		speed is the speed of the incident / reflected waves	B1	[3]

6 (a) (i) 1 total resistance = 0.16 Ω A1